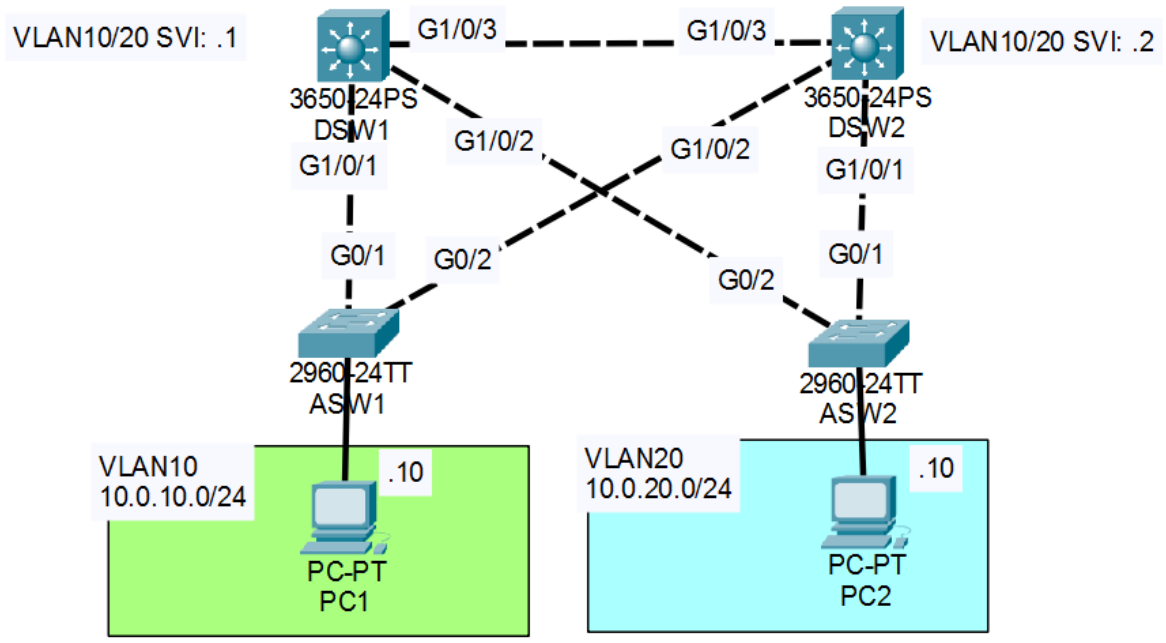
Network Topology:



Instructions and actions:

1. Configure HSRP on DSW1/DSW2, and ensure sychronization with STP.
   In VLAN 10:
   -DSW1 is HSRP active/STP root
   -DSW2 is HSRP standby/STP secondary root

   In VLAN 20:
   -DSW2 is HSRP active/STP root
   -DSW1 is HSRP standby/STP secondary root

DSW1 configurations –

```
DSW1>en
DSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW1(config)#int vlan 10
DSW1(config-if)#standby 1 ip 10.0.10.254
DSW1(config-if)#standby 1 priority 200
DSW1(config-if)#standby 1 preempt
DSW1(config-if)#int vlan 20
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active

DSW1(config-if)#standby 2 ip 10.0.20.254
DSW1(config-if)#standby 2 priority 50
DSW1(config-if)#standby 2 preempt
DSW1(config-if)#
%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

DSW1(config-if)#exit
DSW1(config)#do sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C912.B090
             Cost        4
             Port        3(GigabitEthernet1/0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000C.856A.50BD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Root FWD 4         128.3    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0001.C912.B090
             Cost        4
             Port        3(GigabitEthernet1/0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     000C.856A.50BD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Root FWD 4         128.3    P2p
```

```
VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    32788
             Address     0001.C912.B090
             Cost        4
             Port        3(GigabitEthernet1/0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     000C.856A.50BD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Root FWD 4         128.3    P2p

DSW1(config)#spanning-tree vlan 10 root primary
DSW1(config)#spanning-tree vlan 20 root secondary
DSW1(config)#do sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.C912.B090
             Cost        4
             Port        3(GigabitEthernet1/0/3)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000C.856A.50BD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Root FWD 4         128.3    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     000C.856A.50BD
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     000C.856A.50BD
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Desg FWD 4         128.3    P2p

 --More-- |
```

DSW2 configurations –

```
DSW2>en
DSW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW2(config)#int vlan 10
DSW2(config-if)#standby 1 ip 10.0.10.254
DSW2(config-if)#standby 1 priority 50
DSW2(config-if)#standby 1 preempt
DSW2(config-if)#int vlan 20
DSW2(config-if)#standby 2 i
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

% Ambiguous command: "standby 2 i"
DSW2(config-if)#standby 2 ip 10.0.20.254
DSW2(config-if)#standby 2 priority 200
DSW2(config-if)#standby 2 preempt
DSW2(config-if)#
 %HSRP-6-STATECHANGE: Vlan20 Grp 2 state Standby -> Active

DSW2(config-if)#exit
DSW2(config)#spanning-tree vlan 10 root secondary
DSW2(config)#spanning-tree vlan 20 root primary
DSW2(config)#
```

*Mistake – I had configured the HSRP IPs the same as SVIs on DSW1 and DSW2. That must be different; otherwise, the HSRP protocol will confuse a real IP address with a virtual IP address.

Before checking, use the “ip routing” command in the Global Configuration mode of DSW1 and DSW2.

PC1 –

```
C:\>ping 10.0.20.10

Pinging 10.0.20.10 with 32 bytes of data:

Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 10.0.20.10

Tracing route to 10.0.20.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.10.1
  2   0 ms      0 ms      0 ms      10.0.20.10

Trace complete.
```

PC2 –

```
C:\>ping 10.0.10.10

Pinging 10.0.10.10 with 32 bytes of data:

Reply from 10.0.10.10: bytes=32 time<1ms TTL=127
Reply from 10.0.10.10: bytes=32 time=1ms TTL=127
Reply from 10.0.10.10: bytes=32 time<1ms TTL=127
Reply from 10.0.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.0.10.10

Tracing route to 10.0.10.10 over a maximum of 30 hops:

  1   0 ms      0 ms      1 ms      10.0.20.2
  2   0 ms      0 ms      0 ms      10.0.10.10

Trace complete.
```